



Solving complex linear equations involving brackets

Task A: Solving equations with multiple sets of brackets

1) Fill in the blanks to complete these two equations.

a) $2(5x + 1) + 5(x - 7) = 27$

$$[\] + 2 + 5x - [\] = 27$$

$$15x - [\] = 27$$

$$15x = [\]$$

$$x = [\]$$

b) $4(2x + 6) + 7(4 - x) = 2(2x - 7)$

$$[\] + 24 + 28[\] = [\] - 14$$

$$[\] + 52 = [\] - 14$$

$$52 = [\] - 14$$

$$66 = [\]$$

$$x = [\]$$

2) Each solution contains an error. Identify it and write a sentence explaining why it is wrong.

a) $2(7x + 4) + 3x = 88$

$$14x + 8 + 6x = 88$$

$$20x + 8 = 88$$

$$20x = 80$$

$$x = 4$$

b) $3(2y + 7) - 5(y + 1) = 7(3y - 2)$

$$6y + 21 - 5y + 5 = 21y - 14$$

$$y + 26 = 21y - 14$$

$$26 = 20y - 14$$

$$40 = 20y$$

$$y = 2$$

3) Solve:

a) $2(7x - 3) + 4(x + 9) = 12$

b) $3(2x + 6) + 5(x + 1) = 9(x + 3)$

c) $4(2a + 7) = 12a - 6(a + 8)$

d) $7(3x - 4) + 8(2 - 5x) = 9(x + 6)$

Task B: Simplifying before solving

1) Solve these equations twice. Firstly by expanding first, secondly by simplifying before expanding.

Write a sentence to compare the efficiency.

a) $9(x + 4) + 2(x + 4) = 99$

$$9(x + 4) + 2(x + 4) = 99$$

b) $7 + 5(x - 3) = 8(x - 3) - 2(x - 3)$

$$7 + 5(x - 3) = 8(x - 3) - 2(x - 3)$$

2) Solve:

a) $4x + 24 + 9(x + 6) = 24 + 11(x + 6)$

b) $8(y + 5) = 13 - \frac{y + 5}{2}$